

Diabetes Mellitus

1. Diabetes mellitus is a clinical syndrome characterised by hyperglycaemia caused by absolute or relative deficiency of insulin.
2. Hyperglycaemia has many causes but is most commonly due to type I or type 2 diabetes.
3. Lack of insulin affects the metabolism of carbohydrate, protein and fat, and can cause significant disturbance of water and electrolyte homeostasis
4. Death may result from acute metabolic complication
5. Longstanding metabolic derangement is associated with functional and structural changes in many organs, particularly those of the vascular system, which lead to the clinical 'complications' of diabetes. These characteristically affect the eyes, the kidney and the nervous system.
6. Hyperglycaemia represents an independent risk factor for disease of both small and large blood vessels.
7. The incidences of both type 1 and type 2 diabetes are rising; it is estimated that, in the year 2000, 171 million people had diabetes, and this is expected to double by 2030.
8. This global pandemic principally involves type 2 diabetes and is associated with greater longevity, obesity, unsatisfactory diet, sedentary lifestyle and increasing urbanisation.
9. Many cases of type 2 diabetes remain undetected.
10. The prevalence of both types of diabetes varies considerably around the world, and is related to differences in genetic and environmental factors.

Comparative features of Type 1 and Type 2 Diabetes

	Type 1	Type 2
Typical age at onset	< 40yrs	> 50yrs
Duration of symptoms	Weeks	Months to years
Body weight	Normal or low	Obese
Ketonuria	Yes	No
Rapid death without treatment with insulin	Yes	No
Autoantibodies	Yes	No
Diabetic complications at diagnosis	No	25%
Family history of diabetes	Uncommon	Common
Other autoimmune disease	Common	Uncommon

Aetiology and pathogenesis of diabetes mellitus

1. In both of the common types of diabetes, environmental factor interact with genetic susceptibility to determine which people develop the clinical syndrome and the timing of its onset.
2. However, the underlying genes, precipitating environmental factors and pathophysiology differ substantially between Type 1 and Type 2 diabetes.
3. Type 1 diabetes was previously termed 'insulin-dependent diabetes mellitus' (IDDM) and, is invariably associated with profound insulin deficiency requiring replacement therapy.
4. Type 2 diabetes was previously termed 'non-insulin-dependent diabetes mellitus' (NIDDM) because patients retain the capacity to secrete some insulin but exhibit impaired sensitivity

to insulin (insulin resistance) and initially can usually be treated without insulin replacement therapy.

5. However, 20% or more of patients with type 2 diabetes will ultimately develop profound insulin deficiency requiring replacement therapy so that IDDM and NIDDM were misnomers.

Type 1 diabetes mellitus Pathology

Type 1 diabetes is a T cell-mediated autoimmune disease

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involving destruction of the insulin-secreting β cells in the pancreatic islets which takes place over many years.

↓
Hyperglycaemia accompanied by the classical symptoms of diabetes occurs only when 70-90% of β cells have been destroyed.

↓
'Insulinitis' infiltration of the islets with mononuclear cells containing activated macrophages, helper cytotoxic and suppressor T lymphocytes, natural killer cells and B lymphocytes

↓
Type 1 diabetes mellitus

↓
Hyperglycaemia

↓
Leads to glycosuria and dehydration

↓
Which in turn induces secondary hyperaldosteronism

↓
Unrestrained lipolysis and proteolysis result in weight loss, increased gluconeogenesis and ketogenesis.

↓
Ketoacidosis occurs when generation of ketone bodies exceeds the capacity for their metabolism.

↓
Elevated blood H^+ ions drive K^+ out of the intracellular compartment, while secondary hyperaldosteronism encourages urinary

loss of K^+ .

↓
Hyperglycaemic symptoms (thirst, polyuria, nocturia and fatigue), infections and weight loss

↓
May have developed ketoacidosis

Aetiology

A. Genetic predisposition

1. Genetic factors account for about one-third of the susceptibility to type 1 diabetes, the inheritance of which is polygenic
2. Over 20 different regions of the human genome show some linkage with type 1 diabetes but most interest has focused on the human leucocyte antigen (HLA) region within the major histocompatibility complex on the short arm of chromosome 6; this locus is designated IDDM 1.

B. Environmental factors : Although genetic susceptibility appears to be a prerequisite for type 1 diabetes, the concordance rate between monozygotic twins is less than 40% and environmental factors have an important role in promoting clinical expression of the disease. Although hypotheses abound, the nature of these environmental factors is uncertain.

Clinical features

- | | |
|-----------------------------------|----------------|
| 1. Thirst, | 2. Polyuria |
| 3. Nocturia | 4. Fatigue |
| 5. Infections | 6. Weight loss |
| 7. Ketoacidosis as a complication | |

Type 2 diabetes mellitus Pathology

Type 2 diabetes is a more complex condition

↓
Because there is a combination of resistance to the actions of insulin in liver and muscle together with impaired pancreatic β cell function

↓
Leading to 'relative' insulin deficiency.

↓
Insulin resistance appears to come first,

↓

Leads to elevated insulin secretion in order to maintain normal blood glucose levels.



In susceptible individuals the pancreatic β cells are unable to sustain the increased demand for insulin



A slowly progressive insulin deficiency develops.



Type 2 diabetes mellitus



Intercurrent illness, e.g. with infections



increases the production of counter-regulatory hormones such as cortisol, growth hormone and catecholamines.



This can precipitate an acute exacerbation of insulin resistance and insulin deficiency.



Result in more severe hyperglycaemia and



Hyperosmolar nonketotic coma

Risk factors

1. Central (visceral) obesity
2. Hypertension
3. Dyslipidaemia (characterised by elevated levels of small dense low-density lipoprotein (LDL) cholesterol and triglycerides, and a low level of high-density lipoprotein (HDL) cholesterol)

Aetiology

A. Genetic predisposition :

1. Genetic factors are important in type 2 diabetes, as shown by marked differences in susceptibility in different ethnic groups and by studies in monozygotic twins where concordance rates for type 2 diabetes approach 100%.
2. Many genes are involved and the chance of developing diabetes is also influenced very powerfully by environmental factors
3. Genome-wide association studies have

identified over 20 genes or gene regions that are associated with type 2 diabetes, each exerting a small effect. The largest effect is seen with variation in *TCF7L2*.

B. Environmental and other risk factors

Diet and obesity :

1. Type 2 diabetes is associated with overeating, especially when combined with obesity and underactivity.
2. Middle-aged people with diabetes eat significantly more and are fatter and less active than their non-diabetic siblings.
3. The risk of developing type 2 diabetes increases tenfold in people with a body mass index (BMI) $> 30 \text{ kg/m}^2$
4. Obesity probably acts as a diabetogenic factor (through increasing resistance to the action of insulin) only in those who are genetically predisposed both to insulin resistance and to β cell failure.
5. Sweet foods rich in refined carbohydrate consumed frequently may increase the demand for insulin secretion, while high-fat foods may increase FFAs and exacerbate insulin resistance.

Age

Type 2 diabetes is more common in the middle-aged and elderly

Clinical features of Type 2 DM/ hyperglycaemia :

- Thirst, dry mouth
- Polyuria
- Nocturia
- Tiredness, fatigue, lethargy
- Noticeable change in weight (usually weight loss)
- Blurring of vision
- Pruritus vulvae, balanitis (genital candidiasis)
- Nausea; headache
- Hyperphagia; predilection for sweet foods
- Mood change, irritability, difficulty in concentrating, apathy

Pregnancy

The term 'gestational diabetes' refers to hyperglycaemia occurring for the first time during pregnancy. Repeated pregnancy increases the likelihood of developing irreversible diabetes, particularly in obese women; 80% of women with gestational diabetes ultimately develop permanent diabetes.

Other forms of diabetes

1. Some acquired disorders, notably other endocrine diseases such as acromegaly or Cushing's syndrome, can precipitate type 2 diabetes in susceptible individuals.
2. A number of unusual genetic diseases are associated with diabetes.
3. In rare families, diabetes is caused by single gene defects with autosomal dominant inheritance. These subtypes constitute less than 5% of all cases of diabetes and typically present as 'maturity-onset diabetes of the young' (MODY), i.e. non-insulin-requiring diabetes presenting before the age of 25 years.

Aetiological classification of diabetes mellitus

A. Type 1 diabetes

- Immune-mediated
- Idiopathic

B. Type 2 diabetes

C. Other specific types

- Genetic defects of β cell function
- Genetic defects of insulin action (e.g. leprechaunism, lipodystrophies)
- Pancreatic disease (e.g. pancreatitis, pancreatectomy, neoplastic disease, cystic fibrosis, haemochromatosis, fibrocalculous pancreatopathy)
- Excess endogenous production of hormonal antagonists to insulin (e.g. growth hormone-acromegaly; glucocorticoids-Cushing's syndrome; glucagonoma; catecholamines-phaeochromocytoma; thyroid hormones-thyrotoxicosis)
- Drug-induced (e.g. corticosteroids, thiazide diuretics, phenytoin)

- Viral infections (e.g. congenital rubella, mumps, Coxsackie virus B)
- Uncommon forms of immune-mediated diabetes
- Associated with genetic syndromes (e.g. Down's syndrome; Klinefelter's syndrome; Turner's syndrome; DIDMOAD (Wolfram's syndrome) -diabetes insipidus, diabetes mellitus, optic atrophy, nerve deafness; Friedreich's ataxia; myotonic dystrophy)

D. Gestational diabetes

Investigations

I. Urine testing

A. Glucose

1. Testing the urine for glucose with dipsticks is a common screening procedure for detecting diabetes.
2. If possible, testing should be performed on urine passed 1-2 hours after a meal to maximise sensitivity.
3. Glycosuria always warrants further assessment by blood testing

B. Ketones

1. Ketone bodies can be identified by the nitroprusside reaction, which measures acetoacetate, using either tablets or dipsticks.
2. Ketonuria is therefore not pathognomonic of diabetes but, if associated with glycosuria, the diagnosis of diabetes is highly likely.
3. In diabetic ketoacidosis, ketones can also be detected in plasma using dipsticks.

C. Protein

1. Microalbuminuria or proteinuria, in the absence of urinary tract infection, is an important indicator of the development of diabetic nephropathy and/or increased risk of macrovascular disease

II. Blood testing

Glucose-

1. Glucose concentrations are lower in venous than in arterial or capillary (fingerprick) blood.
2. Whole blood glucose concentrations are lower

than plasma concentrations because red blood cells contain relatively little glucose.

3. In general, venous plasma values are the most reliable for diagnostic purposes

Fasting plasma glucose ≥ 7.0 mmol/L (126 mg/dL)

Random plasma glucose ≥ 11.1 mmol/L (200mg/dL)

Indications for oral glucose tolerance test

- Fasting plasma glucose 6.1-7.0mmol/L (110-126mg/dL)
- Random plasma glucose 7.8-11.0 mmol/L (140-198 mg/dL)

How to perform an oral glucose tolerance test (OGTT) Preparation before the test

- Unrestricted carbohydrate diet for 3 days
- Fasted overnight for at least 8 hrs
- Rest for 30 mins
- Remain seated for the duration of the test, with no smoking

Sampling

Plasma glucose is measured before and 2 hrs after a 75 g oral glucose drink

Interpretation (venous plasma glucose)

	Fasting	2 hrs after glucose load
Fasting	6.1-6.9mmol/L	< 7.8mmol/L
Hyperglycaemia	(110-125mg/dL)	(< 140mg/dL)
Impaired glucose tolerance	< 7.0 mmol/L (< 126 mg/dL)	7.8-11.0 mmol/L (140-199 mg/dL)
Diabetes	≥ 7.0 mmol/L (≥ 126 mg/dL)	≥ 11.1 mmol/L (≥ 200 mg/dL)

Diagnosis of diabetes

- Patient complains of symptoms suggesting diabetes :
- Test urine for glucose and ketones
- Measure random or fasting venous blood glucose. Diagnosis confirmed by:

Fasting plasma glucose ≥ 7.0 mmol/L (126 mg/dL)

Random plasma glucose ≥ 11.1 mmol/L (200mg/dL)

Indications for oral glucose tolerance test (see Box 21.10)

- Fasting plasma glucose 6.1-7.0mmol/L (110-126mg/dL)
- Random plasma glucose 7.8-11.0 mmol/L (140-198 mg/dL)

III. Glycated haemoglobin

1. Glycated haemoglobin provides an accurate and objective measure of glycaemic control over a period of weeks to months.
2. This allows assessment of glycaemic control by repeated measurements every few months in patients with known diabetes, but current assays may not be sufficiently sensitive to make a diagnosis of diabetes and are usually within the normal range in patients with impaired glucose tolerance.
3. Although HbA_{1c} concentration reflects the integrated blood glucose control over the lifespan of the erythrocyte (120 days), half of the erythrocytes are replaced in 60 days and HbA_{1c} is most sensitive to changes in glycaemic control occurring in the month before measurement.

Acute Pancreatitis

- Pancreatitis is an inflammatory process in which pancreatic enzymes autodigest the gland.
- The gland sometimes heals without any impairment of function or any morphologic changes; this process is known as acute pancreatitis.
- Pancreatitis can also recur intermittently, contributing to the functional and morphologic loss of the gland; recurrent attacks are referred to as chronic pancreatitis.

Etiology

Biliary tract disease& Alcohol

- Long-standing alcohol consumption and biliary

stone disease cause most cases of acute pancreatitis, but numerous other etiologies are known.

- As many as 70% of cases of idiopathic pancreatitis are secondary to biliary microlithiasis.

Endoscopic retrograde cholangiopancreatography

Pancreatitis occurring after endoscopic retrograde cholangiopancreatography (ERCP) is probably the third most common type (accounting for approximately 4% of cases).

Trauma

Abdominal trauma (approximately 1.5%) causes an elevation of amylase and lipase levels in 17% of cases and clinical pancreatitis in 5% of cases.

Drugs

Drugs definitely associated with acute pancreatitis include the following:

- Azathioprine
- Sulindac
- Valproic acid,
- Methyldopa
- Furosemide
- Pentamidine
- 5-aminosalicylic acid compounds
- Corticosteroids
- Octreotide
- Sulfonamides
- Tetracycline
- Didanosine
- Estrogens
- 6-Mercaptopurine

Drugs probably associated with acute pancreatitis include the following:

- Chlorothiazide and hydrochlorothiazide
- Methandrostenolone (methandienone)
- Metronidazole
- Phenformin
- Procainamide
- Chlorthalidone
- Combination cancer chemotherapy drugs (especially asparaginase)
- Cimetidine
- Cisplatin
- Nitrofurantoin
- Piroxicam
- Colaspase

- Cytosine arabinoside
- Diphenoxylate
- Ethacrynic acid

Clinical presentation -Signs and symptoms

Symptoms of acute pancreatitis include the following:

- Abdominal pain (cardinal symptom): Characteristically dull, boring, and steady; usually sudden in onset and gradually becoming more severe until reaching a constant ache; most often located in the upper abdomen and may radiate directly through to the back
- Nausea and vomiting, sometimes with anorexia
- Diarrhea

Patients may have a history of the following:

- Recent operative or other invasive procedures
- Family history of hypertriglyceridemia
- Previous biliary colic and binge alcohol consumption (major causes of acute pancreatitis)

The following physical findings may be noted, varying with the severity of the disease:

- Fever (76%) and tachycardia (65%); hypotension
- Abdominal tenderness, muscular guarding (68%), and distention (65%); diminished or absent bowel sounds
- Jaundice (28%)
- Dyspnea (10%); tachypnea; basilar rales, especially in the left lung
- In severe cases, hemodynamic instability (10%) and hematemesis or melena (5%); pale, diaphoretic, and listless appearance
- Occasionally, extremity muscular spasms secondary to hypocalcemia

The following uncommon physical findings are associated with severe necrotizing pancreatitis:

- Cullen sign (bluish discoloration around the umbilicus resulting from hemoperitoneum)
- Grey-Turner sign (reddish-brown discoloration along the flanks resulting from retroperitoneal blood dissecting along tissue planes); more

commonly, patients may have a ruddy erythema in the flanks secondary to extravasated pancreatic exudate

- Erythematous skin nodules, usually no larger than 1 cm and typically located on extensor skin surfaces; polyarthritides

Ancillary investigations symptoms and signs

Laboratory tests are performed to support the clinical impression, such as the following:

- Serum amylase and lipase
- Liver-associated enzymes
- Blood urea nitrogen (BUN), creatinine, and electrolytes
- Blood glucose
- Serum cholesterol and triglyceride
- Complete blood count (CBC) and hematocrit; Neutrophil-Lymphocyte Ratio (NLR)
- C-reactive protein (CRP)
- Arterial blood gas values
- Serum lactic dehydrogenase (LDH) and bicarbonate
- Immunoglobulin G4 (IgG4): elevated in autoimmune pancreatitis

Diagnostic imaging Modalities employed include the following:

- Abdominal radiography (limited role): Kidneys-ureters-bladder (KUB) radiography with the patient upright is primarily performed to detect free air in the abdomen
- Abdominal ultrasonography (most useful initial test in determining the etiology, and is the technique of choice for detecting gallstones)
- Endoscopic ultrasonography (EUS) (used mainly for detection of microlithiasis and periampullary lesions not easily revealed by other methods)
- Abdominal computed tomography (CT) scanning (generally not indicated for patients with mild pancreatitis but always indicated for those with severe acute pancreatitis)
- Endoscopic retrograde cholangiopancreatography (ERCP); to be used with extreme caution in this disease and never as a first-line diagnostic tool

- Magnetic resonance cholangiopancreatography (MRCP) (not as sensitive as ERCP but safer and noninvasive)

Other diagnostic modalities include the following:

- CT-guided or EUS-guided aspiration and drainage
- Genetic testing

Chronic Pancreatitis

- Chronic pancreatitis is commonly defined as a continuing, chronic, inflammatory process of the pancreas, characterized by irreversible morphologic changes.

Signs and symptoms

- For most patients with chronic pancreatitis, abdominal pain is the presenting symptom. The patient experiences intermittent attacks of severe pain, often in the mid-abdomen or left upper abdomen and occasionally radiating in a bandlike fashion or localized to the midback. The pain may occur either after meals or independently of meals, but it is not fleeting or transient and tends to last at least several hours.
- Other symptoms associated with chronic pancreatitis include diarrhea and weight loss.

MCQs

1. Which criteria is useful for the diagnosis of Diabetes mellitus (DM)
 - a. An fasting plasma glucose (FPG), 7.0 mmol/L (126 mg/dL), a glucose > 11.1 mmol/L (200 mg/dL) 2 h after an oral glucose challenge, or an A1C 6.5% .
 - b. A random plasma glucose concentration 11.1 mmol/L (200 mg/dL) accompanied by classic symptoms of DM (polyuria, polydipsia, weight loss)
 - c. both
 - d. Symptoms of diabetes plus a glucose > 11.1 mmol/L (200 mg/dL) 2 h after an oral glucose challenge.

2. Venous blood glucose is about — less than capillary blood due to tissue utilization

- a. 9 mg/dl b. 10 mg/dl
- c. 7 mg/dl d. 11 mg/dl

3. A faint blue discoloration around the umbilicus (Cullen's sign) may occur as the result of hemoperitoneum, and a blue-red-purple or green-brown discoloration of the flanks (Turner's sign) reflects tissue catabolism of haemoglobin indicates —

- a. Mild Acute Pancreatitis
- b. Severe Acute Pancreatitis
- c. Chronic Pancreatitis
- d. Severe necrotizing pancreatitis

Answer

1.	c	2.	c	3.	d
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